Muscle Memory Acquisition in Virtual Reality vs. Real-World Environments

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Abstract

Muscle memory plays a pivotal role in many aspects of daily life and professional practice. This study investigates whether individuals can acquire fine motor muscle memory skills as effectively in a Virtual Reality environment as they can through equivalent real-world practice. Participants are split into two groups: one learning a pattern-drawing task in VR and one learning the same task in the real world. Performance is evaluated based on accuracy and completion time. This research situates itself within the field of human-computer interaction, examining how immersive interface design and controller input influence motor skill learning.

CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; *Virtual reality*.

Keywords

Virtual Reality, muscle memory, motor skill learning, HCI, pattern drawing

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1 Introduction

Muscle memory plays a pivotal role in every aspect of our lives. Not only do we rely on muscle memory to perform day-to-day tasks such as driving a car or washing dishes, we also rely on it in our careers. Surgeons need to have muscle memory when performing high-risk surgeries. Musicians rely on muscle memory to perform complex pieces. These skills are often learned in the field and through experience, which involves a level of risk, resources, and manpower.

If we were able to simulate these environments in a virtual setting, we could create safer learning spaces for students. In these spaces there would be more cushion for mistakes, and resources would be virtual, leading to less need for physical supplies. By creating realistic training environments in Virtual Reality, we can help students build muscle memory in a unique and engaging environment.

The central question of this research is whether someone can pick up a fine motor muscle memory skill as quickly and accurately in Virtual Reality as they can by practicing that same skill in the real world. This could have many implications for the use of Virtual Reality in career training, education, and day-to-day life.

2 Related Work

Previous Virtual Reality studies have demonstrated its effectiveness in applications ranging from complex sports movements [3] to neurorehabilitation [2]. Kampiotis and Theodorakou further note that Virtual Reality has been applied in diverse training contexts, including the instruction of military physicians, naval craft operators, and car drivers [1]. Early implementations of VR-based training date back to the 1990s, with notable studies designed to train pilots using virtual simulations. VR lends itself especially well to motor skill acquisition due to its immersive and interactive nature.

There are many input tools created to support immersion that require motor movement, including head-mounted displays, body gloves, and data gloves. All of these tools can be used to enhance the sensory experience for users and could potentially strengthen the learning of muscle memory movements. Although access to such tools was limited for this study, the use of controllers still lends itself well to simulating fine motor movement similar to that of holding a whiteboard marker. A head-mounted display is also used to provide users with a more immersive, classroom-like experience.

A particular study conducted by Watson and Livingston examined the use of different mixed reality displays, including Virtual Reality, in the learning of motor skills [4]. This study was especially helpful in designing the present experiment. They identified several important confounding factors, including the importance of tools not interfering with the learning experience, avoiding pulling participants out of the virtual experience, and effectively modeling movements for participants throughout the study.

While prior research has demonstrated the effectiveness of Virtual Reality in training contexts, fewer studies directly compare motor memory acquisition in immersive VR environments to equivalent real-world practice using identical fine motor tasks. This project situates itself within the field of human-computer interaction by examining how interface design, controller input, and immersive visual feedback influence motor skill learning.

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3 Methodology

3.1 Task Design

The purpose of this study is to compare how well participants learn a muscle memory skill in Virtual Reality versus the real world. In order to ensure the task is transferable between both environments, a simple pattern-drawing exercise was designed, inspired by the pattern lock software found on smartphones. Participants connect dots arranged in a grid using lines, forming a specific pattern. The task is intuitive enough that the tools used do not distract from the skill being tested, while the pattern itself can be made complex enough to challenge participants.

3.2 Virtual Reality Environment

The Virtual Reality environment for this project is created using Unity. Participants use a Quest 3 headset. The virtual environment is modeled after a classroom. A demo video demonstrating the pattern is projected on a board at the front of the virtual room. Participants then replicate the pattern on a virtual whiteboard next to the projected video using the VR controller. A whiteboard view was chosen because it avoids requiring the participant to look and draw downward, making the interaction more natural. Dots are displayed in a grid pattern on the whiteboard, and some line-snapping functionality is included to make drawing feel more fluid. A timer alerts the participant when to start and stop, but is not displayed, to match the real-world condition. Pattern recognition may also be implemented so participants can receive feedback entirely within the virtual environment.

3.3 Real-World Condition

To mirror the VR condition, participants in the real-world group watch the same demo video on a physical screen. They are given a whiteboard with the dots pre-drawn in marker and connect the dots using a whiteboard marker, replicating the same pattern. The pacing of the experiment is managed by the researcher.

3.4 Experimental Design

Participants are split into two groups: one group learns the pattern in the real world, and the other learns using the Virtual Reality environment. Splitting participants into two groups controls for the confounding variable of pattern complexity. If both groups learn different patterns, results would reflect pattern difficulty rather than the effect of the learning environment. The independent variable is the learning environment (VR versus real-world). The dependent variables are accuracy of pattern replication and time required to complete the pattern during the final timed test.

In both conditions, participants begin by observing the pattern being drawn in the demo video. They then attempt to replicate it. After the first attempt, participants watch the video again, verbally identify what they missed, and try again. This continues until the participant succeeds. Once successful, the participant is timed on three consecutive attempts. At the conclusion of the study, all participants are given a piece of paper and asked to draw the pattern from memory, and are timed. This final test is the primary measure used to collect findings: accuracy (correct or incorrect) and completion time.

3.5 Deliverables

At the end of this project, results from both groups will be compared using the accuracy and timing data from the final test. Charts will be created to visualize both variables and show the relationship between learning environment and performance. These results will be interpreted in the context of existing research in this area.

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